

INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

- Chapters 1 & 2

Origins of the term “Engineer”

- The words “engineer” and “ingenious” both came from the Latin “**ingenium**”, which means a talent, natural capacity, or clever, innovative, creative, good for invention, etc.

Early developments of Engineering

- Science: concerned with the quest of basic knowledge
- Engineering: concerned with the application of scientific knowledge to the solution of problems→ quest of “better life”
- Engineering and science work hand-in-hand
 - Science receives feedback from Engineering (where knowledge is needed?)
 - Great Wall of China, Roman construction projects (Engineering)
 - The inclined plane, the bow, the wheel, the water wheel (Science)
 - Fundamental to all Engineering developments: the advances made in mathematics

Differences between Science and Engineering

- Scientific knowledge is established from observing the world, hypothesizing conjectures with reasoning and logic, to be validated or nullified by controlled physical experiments
 - Scientific conjectures: derived with valid and sophisticated reasoning, consistent with all existing knowledge
 - Scientific theories: conjectures verified through physical experiments

Differences between Science and Engineering

- Engineering is to use/apply established scientific knowledge or scientific theories to the solution of real world problems involving materials, power, structure and systems to make life better for human beings:
 - Food, shelter, movement, communications, environment, etc.

Science has many disciplines and so has Engineering

- Science
 - Physics
 - Chemistry
 - Medical Science
 - Environmental Science
 - Engineering Science
 - Computer Science
 - Animal Science
 - Social Science
 - Others

Science has many disciplines and so has Engineering

- Engineering:
 - Mechanical Engineering
 - Aerospace Engineering
 - Industrial Engineering
 - Building/Civil Engineering
 - Environmental Engineering
 - Electrical Engineering
 - Computer Engineering
 - Chemical Engineering
 - Biomedical Engineering

Engineering: Historical Perspective

- Early engineering marvels
 - Roman Roads
 - Great Wall of China
 - Egyptian Pyramids
 - Many others

Modern Era of Engineering

- Missing early development: behavioral science
 - Important to Industrial Engineers
 - Systems designed by IE involve people as basic components
- Since 1750 (modern era) civil, military, mechanical, electrical, chemical, and Industrial Engineering
 - Increased size and complexity of manufacturing units after rapid technological innovations (mass production)
 - Interchangeability of parts
 - Specialization of labor
 - The need for better management systems
- Developments following World War II: nuclear, electronic, aeronautical, and computer engineering
- Space age: astronautical Engineering
- Environmental concerns: Environmental and bioengineering

Engineers Change the World

- Observe how the world works
- Apply scientific knowledge to improve it
- Implement the ideas
- Some may succeed
- Many will fail
- Lessons learned
- Ultimately, people will have longer and happier life

Engineering as a Profession

- A regulated profession
- Provincial Associations
- National organization: Engineers Canada

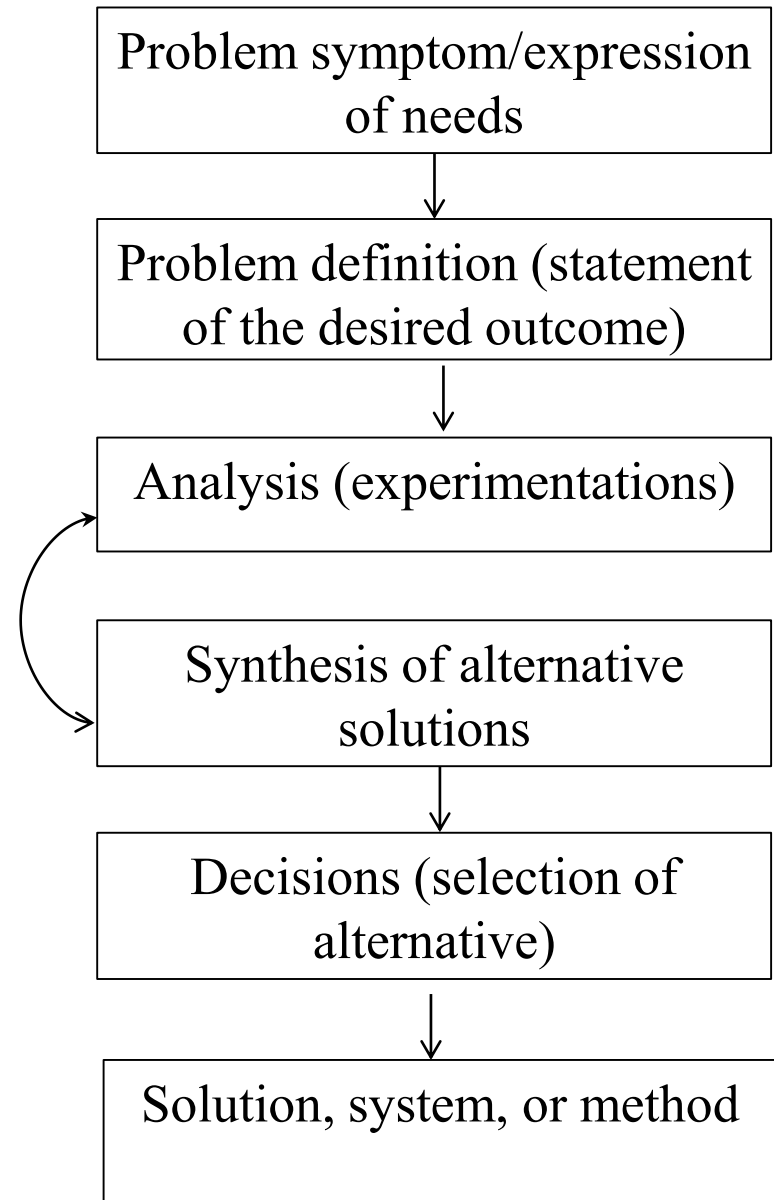
<https://engineerscanada.ca/>

Practice of Engineering

The "practice of engineering" means any act of planning, designing, composing, evaluating, advising, reporting, directing or supervising, or managing any of the foregoing, that requires the application of engineering principles and that concerns the safeguarding of life, health, property, economic interests, the public welfare or the environment.

Engineering process

- Design of systems
 - Analysis
 - Resolving something into basic elements
 - Existing system
 - Synthesis
 - Combining elements into a whole
 - New system



Professional ethics for Engineers

- Engineers involved in decisions with profound impact on society
 - Design of particular devices involve the safety of the user
 - Design of processes affects the environment
 - The design and location of a factory affects the community
- Engineers have a tremendous responsibility to protect the public welfare
 - They must engage in trade-off between costs and public welfare (e.g., safety)
- They are caught in a controversy between their employer and the public
 - A cheap design may mean an unsafe product
- Professional licensing (e.g., OIQ)

Chronology of Industrial Engineering

- Emerged as a profession as a result of the industrial revolution
 - People who could plan, organize, and direct operations of large complex systems
 - Increase efficiency and effectiveness of operations
- Division of labor and harmonious labor relationship (Charles Babbage, 1800's)
- Mass production and assembly lines (H. Ford) enabled by:
 - Interchangeable manufacture
 - Machines that could be operated by labors with minimum level of training
- Job analysis design for maximum efficiency (F.W. Taylor)
- Motion analysis and Time Study (F.B. Gilberth)
- Human factors (L. Gilberth)
- Gantt chart (H.L. Gantt)
- Quality Control (Shewhart)

Chronology of Industrial Engineering

- Mass production
 - Productivity and work standardization
 - Human relations
 - Gantt charts
 - Statistical quality control
 - Inventory planning and control
 - Facility layout and material handling
 - Operations Research
 - Scientific research on operations
 - First applications in military
 - **Business and data analytics**
- } Scientific management

Definition of the Discipline

Industrial Engineering is concerned with the design, improvement, and installation of integrated systems of people, materials, information, equipment, and energy. It draws upon specialized knowledge and skill in the mathematical, physical, and social sciences together with the principles and methods of engineering analysis and design to specify, predict, and evaluate the results to be obtained from such systems.

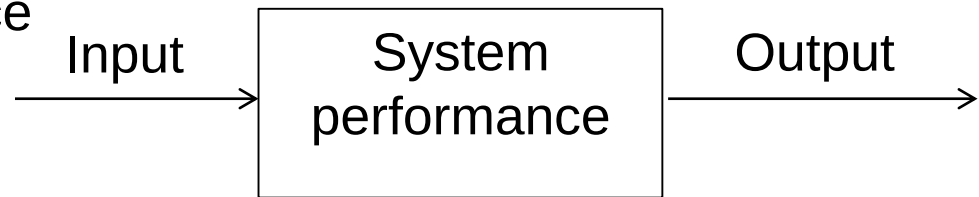
Industrial and system Engineering

- What distinguishes IE from other engineering disciplines?
 - Design of industrial systems
 - All types of organizations, not just manufacturing
 - Design systems at two levels:
 - Human activity systems (physical workplace)
 - Manufacturing process, materials, machines, methods, layout, material handling, storage space, data recording, maintenance procedures, safety procedures
 - Management control systems (procedures for planning, measuring, and controlling activities)
 - Forecasting procedure, budgeting, incentive plans, recruiting/training, material requirement plan, inventory control, production scheduling, quality control, overall information system, etc.
- Improving the performance of a small part of the system (what we actually do in practice)

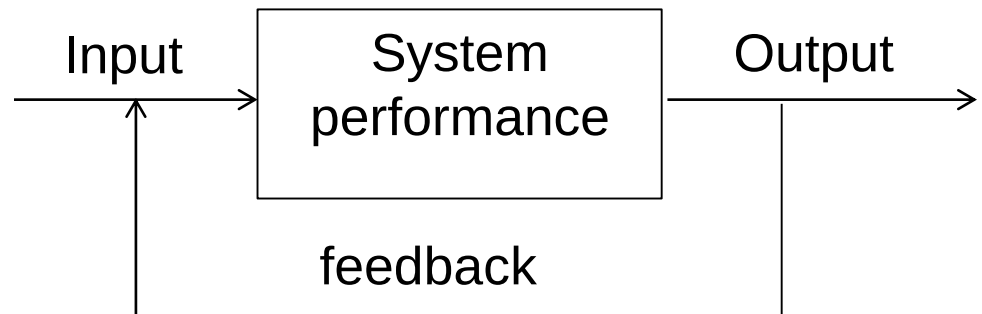
The nature of systems

- A set of components , related by some form of interaction, which act together to achieve some objectives
- All systems exist to perform some function
- Open-loop system

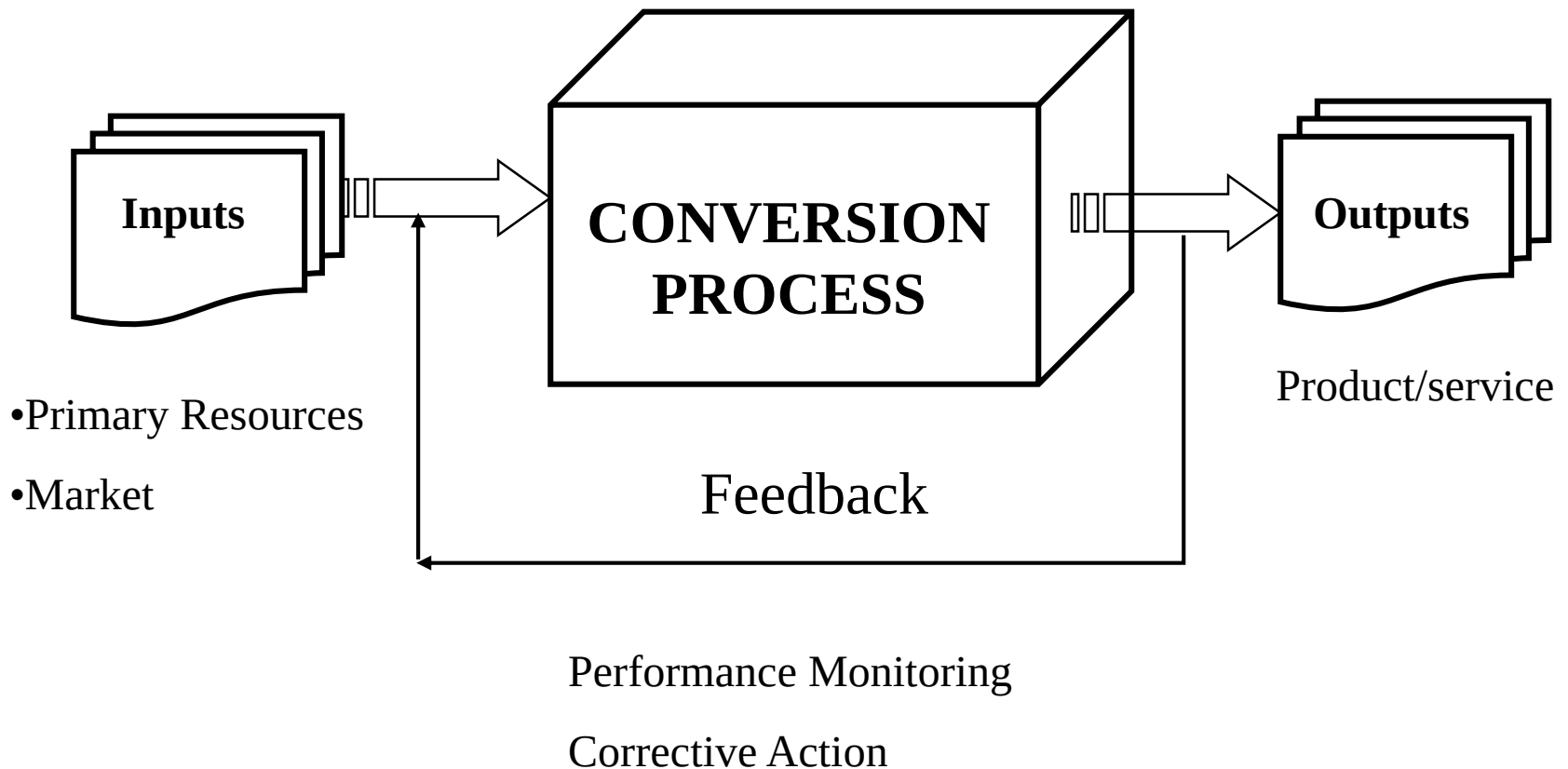
- Not aware of its performance
- No mean to provide control
- Auto without a driver



- Closed-loop system
- Aware of and influenced by its past performance
- Auto with a driver



Components of a Production System



Decision-Making

- Strategic decisions: what, how, where?
 - Products, processes, facilities
- Tactical decisions: how much, when?
 - Production planning to meet demand
 - Inventory planning
 - Maintenance planning
- Control decisions
 - Planning and controlling operations

IE Solve Problems

- With a BROAD engineering foundation and overall approach
- By realizing that systems always contain people
- By being more people oriented in their solutions

INDUSTRIAL ENGINEERS

Work with

PEOPLE

To do things

BETTER, FASTER, SAFER, CHEAPER

Comparing Different Engineering Disciplines

- Mechanical: mainly concerned with products that move (cars, trains, machines, airplanes, etc.)
 - Scientific knowledge: physics/dynamics
 - To be used with/for: metals and other materials
- Civil: mainly concerned with products that do not move (buildings, bridges, roads, etc.)
 - Scientific knowledge: physics/statics
 - To be used with/for: concrete and other materials

Comparing Different Engineering Disciplines

- Electrical: products powered/controlled/connected to electricity (power grid, electric motors, generators, etc.)
 - Scientific knowledge: physics/electro-magnetics
 - To be used with/for: Electrical/electronic components and other materials
- Industrial: products/processes as systems with multiple connected components and operated/managed by people (factories, warehouses, hospitals, schools, banks, etc.)
 - Scientific knowledge: applied math/operations research
 - To be used with/for: people (users and workers, service providers), products and systems

Industrial Engineering Job Opportunities

- Production and Service Industries
 - Manufacturing
 - Health Care/Hospitals
 - Financial Institutions
 - Transportation
 - Postal services
 - Retail
 - Entertainment
 - Government/public services

Industrial Engineering Job Opportunities

- Job Title: Industrial Engineer
- Possible other titles:
 - Process control engineer
 - Quality engineer
 - Optimization engineer
 - System engineer
 - Manufacturing engineer
 - Facility management engineer
 - Process improvement engineer

Industrial Engineers Earn

- Not bad comparing to other engineering disciplines
- Entry-level positions: 50-80K CAD

Advice from an IISE Member

“Successful industrial engineers must possess the ability to communicate effectively, for without it you cannot sell your ideas. You must be able to manage projects and multiple tasks, for without those skills you will be less efficient and of less use to your employer. You must be able to observe others and understand why they are doing what they do, for without that change is an uphill battle.”

Plan for Success

“... about 15 percent of one’ s financial success is due to one’ s technical knowledge and about 85 percent is due to skill in human engineering – to personality and ability to lead people.”

Dale Carnegie ’s Lifetime Plan for Success, pg.9, Galahad Books, 1998.

Current Factors Affecting IE

- Computers, advanced production technology
- Global competition
- Time-based competition
- Supply chain management
- Sustainability movement
- Big data
- Smart cities
- Smart factories
- Large Language modeling (Generative AI)

Industrial Engineering in the Age of AI

- AI is enhancing industrial engineering, not displacing it.
- Industrial engineers are now tasked with designing and managing AI-powered systems to optimize production, reduce waste, and predict equipment failures.
- Ability to work with automation systems and understand the intricacies of the machine learning algorithms that power these technologies.
- Transforming industrial engineering jobs by enabling engineers to make data-driven decisions that improve efficiency and performance.
- By analyzing real-time data from production processes, engineers can identify bottlenecks, predict equipment failures, and optimize supply chains.

Some Useful Web Sites on our Profession

Ordre des ingénieurs du Québec (OIQ)

The **Ordre des ingénieurs du Québec (OIQ)** is the self regulatory body that governs Quebec's 55,000 professional engineers. In Quebec, the OIQ carefully monitors compliance with rules of this trade and with the professional integrity of members, while also overseeing the development of the engineering profession.

(Source: Wikipedia.org)

<http://www.oiq.qc.ca/en/>

Some Useful Web Sites on our Profession

Engineers Canada

- Engineers Canada is the national organization of the 12 provincial and territorial associations that regulate the practice of engineering in Canada. Engineers Canada serves these associations, which are its sole members, by delivering national programs for standards of engineering education, professional qualifications and professional practice.

Some Useful Web Sites on our Profession

- **Canadian Engineering Accreditation Board (CEAB)**

Through the Canadian Engineering Accreditation Board (CEAB), Engineers Canada accredits Canadian undergraduate engineering programs that meet the profession's education standards. Graduates of those programs are deemed by the profession to have the required academic qualifications to be licensed as professional engineers in Canada. (Source: Wikipedia.org)

Some Useful Web Sites on our Profession

- Institute of Industrial and System Engineers (IISE)
<https://www.iise.org/Home/>
- Industrial and systems engineers make things better in any industry — from automobile manufacturing and aerospace, to healthcare, forestry, finance, leisure, and education. The Institute of Industrial and Systems Engineers, founded in 1948, helps its members improve complex organizations around the world and across industries. Throughout their careers, members turn to IISE for tools and connections that provide an integrated and systemic perspective to business challenges. Solving complex problems is our common objective.



Some Useful Web Sites on our Profession

- IISE Concordia
- <https://www.facebook.com/IISEConcordiaChapter/>
- Other information from the department website